

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Paper No. CHB-505: Environmental Chemistry & Nuclear Chemistry

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Environmental Chemistry – Marks: 35)

Question 1. Answer all parts of the following: $[5 \times 2 = 10]$

- (a) Define the terms: tropopause and stratosphere.
- (b) What is meant by a 'sink' in environmental chemistry? Give one example.
- (c) What is organic soot? Write its environmental sources.
- (d) List the major greenhouse gases and their relative contributions to global warming.
- (e) What is the hydrosphere?

Question 2.

- (a) What are the broad categories of water pollutants? What are the important water quality parameters and how do you estimate them? $[8]$
- (b) Discuss the statement: "Ozone acts as a protective shield for living organisms on earth." Explain its depletion by chlorofluorocarbons (CFCs). $[6]$

Question 3. Write detailed notes on any two of the following: $[2 \times 7 = 14]$

- (a) Photochemical Smog formation and its control
- (b) Nitrogen cycle and phosphate cycle in nature
- (c) Arsenic and Mercury poisoning and their toxicological effects
- (d) Municipal wastewater treatment methods

SECTION B

(Nuclear Chemistry – Marks: 35)

Question 4. Answer all parts of the following: [5 × 2 = 10]

- (a) What is a nuclear reaction cross section (σ)? How is it different from the geometric cross section?
- (b) Thermal neutrons interact with ^{235}U in two ways: (i) radiative capture (n, γ) with $\sigma_c = 93 \text{ b}$ and (ii) fission (n, f) with $\sigma_f = 584 \text{ b}$. Explain why they are different.
- (c) What are the tracer applications of radioisotopes in chemistry?
- (d) Differentiate between nuclear fission and nuclear fusion.
- (e) Define half-life and decay constant of a radioactive isotope.

Question 5.

- (a) Discuss the compound nucleus theory of nuclear reactions and outline its salient features. [7]
- (b) Describe the construction and working of the Geiger-Müller counter for the detection of nuclear radiations. [7]

Question 6.

- (a) Discuss the shell model of the nucleus. What are magic numbers and how do they explain nuclear stability? [8]
- (b) Discuss the principles and applications of neutron activation analysis (NAA). [6]

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